Seeing Together: question and success criteria

Smart Glasses-Augmented Spatial Memory for Human-Robot Collaboration

DESIGN GOAL

In an indoor lab, a person's smart glasses and a Go2 robot dog write timestamped observations into one shared spatial memory, so the robot can find an object that moved out of its own view.

KEY CONSTRAINT

Reliable glasses-hub Wi-Fi streaming with spatiotemporal consistency across human-robot views.

WHY IT MATTERS

- Human observations transform outdated robot memory into actionable intelligence for accurate object retrieval.

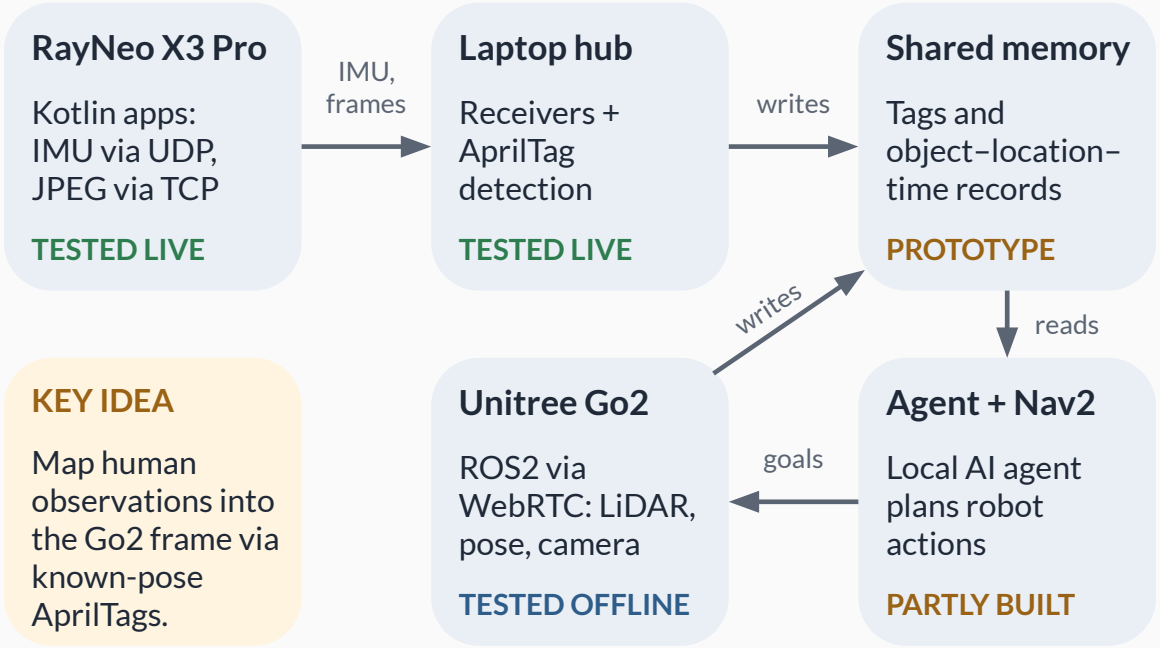


SUCCESS CRITERIA

- **Human:** Reliable glasses streaming (IMU ≥50 Hz, <1% loss; video ≥25 fps, <200 ms, AprilTag 1–3 m)
- **Robot:** Accurate mapping and reconstruction, LLM-guided search
- **Final:** beat Robot-Only on task success

Approach and current implementation

Glasses and robot send observations to a laptop hub that keeps one shared memory



TEST SETTING

- HW/SW: RayNeo X3 Pro, laptop (OpenCV), Go2 + ROS2
- Inputs: live IMU/camera, tags on screen, lab map
- Mode: glasses live; robot: saved-map replay

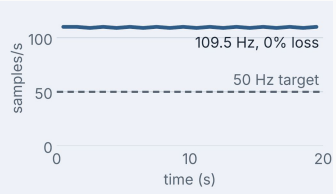
TEAM CONTRIBUTION

- Hyun Kyum: glasses apps, receivers, latency test
- Khang: ROS2 sensing, map projection, agent
- Reused: Camera2, OpenCV, Go2 SDK, ROS2

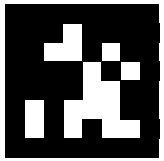
Preliminary evidence and what it means

At 640x480 the glasses stream 30 fps at ~100 ms; larger frames are encoder-bound

Link, frame size	fps	Dropped	Latency med / p95
Wi-Fi, 640x480	29.8	0.8%	106 / 133 ms
Wi-Fi, 1280x720	18.6	38%	135 / 168 ms
Wi-Fi, 1920x1080	9.5	68%	206 / 244 ms
USB, 640x480	29.5	1.5%	96 / 118 ms

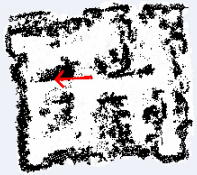


IMU 109.5 Hz
0% loss in 20 s;
survives a Wi-Fi drop



Test tag
tag36h11
ID 0

Tags 18 / 18
Stills at 0.5–3 m, up to ~60°



Robot map
Lab map and robot navigation

OBSERVATION

Glasses: IMU and video criteria met at 640x480 over Wi-Fi; tags detected up to 3 m. USB offered no gain.
Robot: LiDAR map built with robot location and navigation

INTERPRETATION

Glasses: Wi-Fi suffices; above 640x480, on-device encoding limits frame rate.

LIMITATION

No tag pose accuracy yet; at 2 m, 15 cm tags found in 18–93% of frames.

Code (commit 0eede7c):
github.com/hyunkyumkim/seeing-together

Table: preliminary results from the Nov 2 check; 40 s live runs, n = 178–590 samples; AprilTag clock (50 ms steps) + monitor delay.

Evaluation plan and next steps

We compare the shared memory against Robot-Only on finding moved objects

EVALUATION

- Baselines: Robot-Only, Naive Human+Robot
- Primary metric: task success rate, search time
- Test: objects moved in the lab; stress: out of robot view, Wi-Fi drop
- Needed: printed tags, calibration target

MAIN RISK

Tag pose error too large for the map. Fallback: snap to the nearest saved lab place.

SCOPE

Minimum deliverable: the Go2 reaches an object the glasses saw being moved.

NEXT STEPS	Owner	Target
Camera calibration; tag pose error vs tape-measured positions	Hyun Kyum	Nov 17
Live glasses tag and IMU observations into the shared memory	Hyun Kyum	Nov 24
Clearer map and memory construction; LLM-controlled navigation	Khang	Nov 24

FEEDBACK REQUESTED Is a home test room with tape-measured tags acceptable before the lab demo?